Improve constant approximation bound for Offset regret minimization under additive noise

ChatGPT (`gpt-5.6-sol`)*

Solution generated on 2026-08-21

Abstract

Mahdian, Mao, and Wang introduced a minimax-regret model for selecting the largest of several unknown values observed through independent additive noise. They proved that their offset algorithm is a 57000-approximation to the optimal worst-case regret and asked whether a better constant could be obtained. We prove that the same algorithm is a 6508-approximation. The improvement comes from a sharper lower bound for the optimal regret of the anchored instance appearing in their reduction. In particular, a wider pigeonhole interval and a rebalanced pair of indistinguishable instances improve the anchored lower bound to $3b/3800$.

1 The open problem

This paper addresses an open problem posed by Mohammad Mahdian, Jieming Mao, Kangning Wang in *Regret Minimization with Noisy Observations* [1] (Economics and Computation (EC), 2023), Section 8 (Conclusions), page 17 (arXiv:2207.09435v1). The website catalogue entry is problem 13 (`W4383558051_p0`) of the Open Problems in Operations Research collection.

> *For the specific offset mapping θ defined by the minimax threshold rule in the binary case and applied itemwise, prove an improved universal constant-factor guarantee*
>
> $$\mathrm{Reg}(\mathrm{Offset}_\theta) \le c\,\mathrm{OPT}(A)$$
>
> *with some constant c strictly smaller than the currently proved value ($c = 57000$), uniformly over all numbers of items n and all collections of independent mean-zero noise distributions $A = (A_1, \ldots, A_n)$ satisfying that each $\varphi_i(x) = \Pr_{a_i\sim A_i}(|a_i| \ge x)\cdot x$ is bounded on $[0, \infty)$.*

The remainder of this document is the solution produced by the model, reproduced verbatim from the pipeline output.

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*Generated by the `base_solver_non_interactive` pipeline (Codex CLI, non-interactive; reasoning effort `xhigh`; stages A.6 solve → A.8 verify → A.9 write-up; run `run_20260821_045718_068409e3, ec-xhigh`). Independent verifier assessment: APPROVED with progress score 3/3 (full solution). Maintained by the AI in Mathematics & Operations Research project.

2 Introduction

Consider n items with unknown intrinsic values $v_1, \dots, v_n$. Item i is observed through an independent additive noise a_i with known distribution A_i, so that the algorithm observes

$$s_i = v_i + a_i.$$

The intrinsic values are chosen adversarially. The objective is to select one item while minimizing the worst-case expected difference between the largest intrinsic value and the value of the selected item.

Mahdian, Mao, and Wang [2] introduced this model and the algorithm Offset_θ, which selects an item maximizing

$$s_i - \theta_i, \qquad \theta_i = \theta(A_i),$$

where the offset $\theta(A_i)$ is obtained from a one-dimensional minimax problem depending only on A_i. They proved that Offset_θ is a 24-approximation in the binary case and a 57000-approximation in general. They also exhibited instances with approximation ratio arbitrarily close to 2, and explicitly asked whether the upper bound could be improved.

Our main result answers that question affirmatively.

Theorem 1. *For every number of items and every collection of independent noise distributions satisfying the bounded-tail assumption of [2],*

$$\mathrm{Reg}(\mathrm{Offset}_\theta; \boldsymbol{A}) \leq 6508\ \mathrm{OPT}(\boldsymbol{A}).$$

The proof retains the binary theorem and the general-to-anchored reduction of [2]. The new ingredient is the following anchored estimate:

$$\mathrm{OPT}(\mathbf{0}, A_2, \dots, A_n) \geq \frac{3}{3800}\, b,$$

where b is the value of the linear program arising in the linearization step of the original proof.

3 Model and notation

Let $N = \{1, \dots, n\}$. Item i has intrinsic value $v_i \in \mathbb{R}$, noise distribution A_i, and observed value $s_i = v_i + a_i$, where the a_i are mutually independent and $a_i \sim A_i$. The known means of the noise distributions may be subtracted from the observations, so the noises may be assumed centered.

For an algorithm $\mathcal{M}$, a collection of noise distributions $\boldsymbol{A} = (A_1, \dots, A_n)$, and a value vector $\boldsymbol{v}$, define

$$\mathrm{Reg}(\mathcal{M}; \boldsymbol{A}, \boldsymbol{v}) = \max_{i \in N} v_i - \mathbb{E}\left[v_{\mathcal{M}(\boldsymbol{s}, \boldsymbol{A})}\right],$$

where the expectation includes both the noise and any internal randomness of $\mathcal{M}$. Its worst-case regret is

$$\mathrm{Reg}(\mathcal{M}; \boldsymbol{A}) = \sup_{\boldsymbol{v} \in \mathbb{R}^n} \mathrm{Reg}(\mathcal{M}; \boldsymbol{A}, \boldsymbol{v}),$$

and the optimal regret is

$$\mathrm{OPT}(\boldsymbol{A}) = \inf_{\mathcal{M}} \mathrm{Reg}(\mathcal{M}; \boldsymbol{A}).$$

Following [2], for a one-dimensional noise distribution D, the offset $\theta(D)$ minimizes

$$\max\left\{\sup_{u>0} u\,\mathbb{P}[D \le t-u],\ \sup_{u<0}(-u)\,\mathbb{P}[D \ge t-u]\right\}$$

over $t \in \mathbb{R}$. The algorithm Offset_θ selects an item in

$$\arg\max_{i\in N}\{s_i - \theta(A_i)\}.$$

A fixed tie-breaking convention is used throughout.

We assume, as in [2], that

$$\sup_{x\ge 0} x\,\mathbb{P}(|a_i| \ge x) < \infty$$

for every i. This ensures that the relevant worst-case regrets and optimization problems are finite.

If μ and ν are measures, we write

$$\mu \succeq c\nu$$

when $\mu(E) \ge c\nu(E)$ for every measurable set E.

4 Ingredients from the original analysis

We use three results from Sections 5–6 of [2].

First, the binary theorem gives

$$\mathrm{Reg}(\mathrm{Offset}_\theta; (D, \mathbf{0})) \le 24\ \mathrm{OPT}(D, \mathbf{0}). \tag{1}$$

Second, the reduction in Section 6.1 separates a general instance into an anchored instance and a binary instance. Applied to an arbitrarily nearly worst value vector and relabelling its highest-valued item as item 1, it yields

$$\mathrm{Reg}(\mathrm{Offset}_\theta; \boldsymbol{A}) \le 2\,\mathrm{Reg}^{\uparrow}(\mathrm{Offset}_\theta; (\mathbf{0}, A_2, \dots, A_n)) + 2\,\mathrm{Reg}(\mathrm{Offset}_\theta; (A_1, \mathbf{0})). \tag{2}$$

Here the upward arrow records the restriction produced by the reduction:

$$\mathrm{Reg}^{\uparrow}(\mathrm{Offset}_\theta; (\mathbf{0}, A_2, \dots, A_n)) := \sup_{\boldsymbol{v}\colon v_1 = \max_j v_j} \mathrm{Reg}(\mathrm{Offset}_\theta; (\mathbf{0}, A_2, \dots, A_n), \boldsymbol{v}).$$

Thus the noiseless item is required to have the largest intrinsic value. After translating all values by a common constant, this is equivalent to imposing

$$v_1 = 0, \qquad v_i \le 0 \quad (i \ge 2).$$

For $i \ge 2$, define

$$p_i(x) := \mathbb{P}[a_i \ge \theta_i + x], \qquad \theta_i := \theta(A_i),$$

and let

$$b := \sup\left\{\sum_{i=2}^{n} x_i p_i(x_i) : x_i \ge 0, \quad \sum_{i=2}^{n} p_i(x_i) \le \frac{1}{2}\right\}. \tag{3}$$

The linearization argument of Section 6.2 gives

$$\mathrm{Reg}^{\uparrow}(\mathrm{Offset}_\theta; (\mathbf{0}, A_2, \dots, A_n)) \le \frac{51}{20}\, b. \tag{4}$$

It remains to obtain a stronger lower bound on the optimal regret in terms of b.

5 The improved anchored lower bound

Theorem 2. *For b defined by (3),*

$$\mathrm{OPT}(\mathbf{0}, A_2, \ldots, A_n) \geq \frac{3}{3800}\, b.$$

Proof. We first assume that the supremum in (3) is attained. The general case is treated at the end of the proof. If $b = 0$, the claim is immediate, so suppose $b > 0$.

Choose an optimizer $(x_2, \ldots, x_n)$, and abbreviate

$$p_i := p_i(x_i) = \mathbb{P}[a_i \geq \theta_i + x_i].$$

Thus

$$\sum_{i=2}^{n} p_i x_i = b, \qquad \sum_{i=2}^{n} p_i \leq \frac{1}{2}. \tag{5}$$

Define

$$I := \{i \in \{2, \ldots, n\} : x_i \geq b\}.$$

Because $x_i < b$ outside I, (5) implies

$$\begin{aligned} \sum_{i \in I} p_i x_i &= b - \sum_{i \notin I} p_i x_i \\ &\geq b - b \sum_{i \notin I} p_i \geq \frac{b}{2}. \end{aligned} \tag{6}$$

5.1 A tail interval from coordinate optimality

Fix $i \in I$. Replacing x_i by $7x_i/5$ cannot violate the probability constraint in (3), since $p_i(x)$ is nonincreasing. Optimality therefore gives

$$\frac{7}{5} x_i\, p_i\!\left(\frac{7}{5} x_i\right) \leq x_i p_i.$$

Consequently,

$$\begin{aligned} \mathbb{P}\left[a_i \in \left[\theta_i + x_i, \theta_i + \frac{7}{5} x_i\right)\right] &= p_i - p_i\left(\frac{7}{5} x_i\right) \\ &\geq \frac{2}{7} p_i. \end{aligned} \tag{7}$$

5.2 Two easy binary cases

We first dispose of two cases in which the binary theorem (1) already gives the desired bound.

Suppose that for some $i \in I$,

$$\mathbb{P}[a_i < \theta_i - 5x_i] \geq \frac{1}{5}.$$

Consider the binary instance consisting of a noiseless item of value zero and noisy item i of value $5x_i$. On the displayed event, Offset_θ fails to select the noisy item. Hence

$$\mathrm{Reg}(\mathrm{Offset}_\theta; (\mathbf{0}, A_i)) \geq 5x_i \cdot \frac{1}{5} = x_i.$$

The binary theorem and the embedding of this binary problem in the full anchored problem give

$$\mathrm{OPT}(\mathbf{0}, A_2, \ldots, A_n) \geq \mathrm{OPT}(\mathbf{0}, A_i) \geq \frac{x_i}{24} \geq \frac{b}{24} > \frac{3b}{3800}.$$

Alternatively, suppose that for some $i \in I$,

$$\mathbb{P}\left[a_i \in \left[\theta_i + \frac{1}{5}x_i, \theta_i + x_i\right)\right] \geq \frac{9}{95}.$$

Give noisy item i intrinsic value $-x_i/5$, while the noiseless item has value zero. On the displayed event, Offset_θ selects the inferior noisy item. Therefore

$$\mathrm{Reg}(\mathrm{Offset}_\theta; (\mathbf{0}, A_i)) \geq \frac{x_i}{5} \cdot \frac{9}{95}.$$

It follows that

$$\mathrm{OPT}(\mathbf{0}, A_2, \ldots, A_n) \geq \frac{1}{24} \cdot \frac{x_i}{5} \cdot \frac{9}{95} = \frac{3x_i}{3800} \geq \frac{3b}{3800}.$$

We may henceforth assume that neither easy case occurs for any $i \in I$.

5.3 A pigeonhole interval

For $i \in I$, consider

$$H_i := \left[\theta_i - 5x_i,\ \theta_i + \frac{1}{5}x_i\right).$$

The complement of H_i is covered by the lower-tail event from the first easy case, the interval from the second easy case, and the event $\{a_i \geq \theta_i + x_i\}$. Since $p_i \leq 1/2$, we obtain

$$\begin{aligned}\mathbb{P}[a_i \in H_i] &\geq 1 - \frac{1}{5} - \frac{9}{95} - \frac{1}{2} \\ &= \frac{39}{190}.\end{aligned} \tag{8}$$

The length of H_i is $26x_i/5$. Partition it into thirteen half-open intervals of length $2x_i/5$. By (8), one of them has mass at least

$$\frac{1}{13} \cdot \frac{39}{190} = \frac{3}{190}.$$

Thus there is a number k_i, with

$$\frac{1}{5} \leq k_i \leq 5,$$

such that

$$B_i := \left[\theta_i - k_i x_i,\ \theta_i + \left(\frac{2}{5} - k_i\right) x_i\right) \tag{9}$$

satisfies

$$\mathbb{P}[a_i \in B_i] \geq \frac{3}{190}. \tag{10}$$

5.4 Common observation components

Let

$$T_i := \mathrm{Unif}\left[-\frac{9}{5}x_i, -x_i\right], \qquad W_i := \mathrm{Unif}\left[\left(k_i - \frac{4}{5}\right)x_i, k_i x_i\right],$$

and define

$$J_i^0 := \left[\theta_i - \frac{2}{5}x_i, \theta_i\right].$$

Convolution with a uniform distribution is absolutely continuous, even when A_i has atoms. For almost every $t \in J_i^0$,

$$f_{T_i+A_i}(t) = \frac{5}{4x_i}\mathbb{P}\left[a_i \in \left[t + x_i, t + \frac{9}{5}x_i\right]\right].$$

For every such t, the interval on the right contains

$$\left[\theta_i + x_i, \theta_i + \frac{7}{5}x_i\right).$$

Using (7),

$$f_{T_i+A_i}(t) \geq \frac{5}{4x_i} \cdot \frac{2}{7}p_i.$$

The density of $\mathrm{Unif}[J_i^0]$ is $5/(2x_i)$, and hence

$$T_i + A_i \succeq \frac{p_i}{7}\,\mathrm{Unif}[J_i^0]. \tag{11}$$

Similarly,

$$f_{W_i+A_i}(t) = \frac{5}{4x_i}\mathbb{P}\left[a_i \in \left[t - k_i x_i,\, t + \left(\frac{4}{5} - k_i\right)x_i\right]\right].$$

For $t \in J_i^0$, this interval contains B_i. Therefore (10) gives

$$f_{W_i+A_i}(t) \geq \frac{5}{4x_i} \cdot \frac{3}{190},$$

and consequently

$$W_i + A_i \succeq \frac{3}{380}\,\mathrm{Unif}[J_i^0]. \tag{12}$$

Shift both value distributions to the right by $4x_i/5$, and put

$$\widetilde{T}_i := \mathrm{Unif}\left[-x_i, -\frac{1}{5}x_i\right], \qquad \widetilde{W}_i := \mathrm{Unif}\left[k_i x_i, \left(k_i + \frac{4}{5}\right)x_i\right].$$

Every value in the support of $\widetilde{T}_i$ is at most $-x_i/5$. Since $k_i \geq 1/5$, every value in the support of $\widetilde{W}_i$ is at least $x_i/5$.

After this shift, (11) and (12) become

$$\widetilde{T}_i + A_i \succeq \frac{p_i}{7}\,\mathrm{Unif}[J_i], \tag{13}$$

$$\widetilde{W}_i + A_i \succeq \frac{3}{380}\,\mathrm{Unif}[J_i], \tag{14}$$

where

$$J_i := \left[\theta_i + \frac{2}{5}x_i,\, \theta_i + \frac{4}{5}x_i\right].$$

5.5 The hard prior

Consider the following prior on intrinsic value vectors.

- With probability
$$w_0 := 1 - \sum_{i \in I} p_i \geq \frac{1}{2},$$
all coordinates $i \in I$ are drawn independently from $\widetilde{T}_i$. Call this the all-typical scenario.

- For each $i \in I$, with probability p_i, coordinate i is instead drawn from $\widetilde{W}_i$, while every $j \in I \setminus \{i\}$ is drawn from $\widetilde{T}_j$. Call this atypical scenario i.

- In every scenario, set $v_1 = 0$ and set $v_j = 0$ for $j \notin I$. Their original noise distributions are retained.

The scenarios have total probability one because $\sum_{i \in I} p_i \leq 1/2$.

Fix any randomized algorithm $\mathcal{M}$. Let q_i be the probability that $\mathcal{M}$ selects item i in the all-typical scenario, and let q'_i be the probability that it selects item i in atypical scenario i.

Define an auxiliary observation experiment in which s_i is uniform on J_i, while all other coordinates have their all-typical distributions. Let r_i be the probability that $\mathcal{M}$ selects item i in this experiment.

The joint observation distributions of all coordinates other than i are identical in these three experiments. Therefore the measure dominations (13) and (14) imply

$$q_i \geq \frac{p_i}{7} r_i \tag{15}$$

and

$$1 - q'_i \geq \frac{3}{380}(1 - r_i). \tag{16}$$

In the all-typical scenario, selecting item i loses at least $x_i/5$, since $v_i \leq -x_i/5$ while $v_1 = 0$. In atypical scenario i, failing to select i also loses at least $x_i/5$, since $v_i \geq x_i/5$ and every other intrinsic value is at most zero. Hence the prior-average regret of $\mathcal{M}$ is at least

$$\begin{aligned} \sum_{i \in I} \frac{x_i}{5} \left(w_0 q_i + p_i(1 - q'_i)\right) &\geq \sum_{i \in I} \frac{x_i}{5} \left(\frac{1}{2} q_i + p_i(1 - q'_i)\right) \\ &\geq \sum_{i \in I} \frac{p_i x_i}{5} \left(\frac{r_i}{14} + \frac{3}{380}(1 - r_i)\right). \end{aligned} \tag{17}$$

Since

$$\frac{3}{380} < \frac{1}{14},$$

the expression in parentheses is at least $3/380$ for every $r_i \in [0, 1]$. Thus, by (6),

$$\begin{aligned} \mathbb{E}_{\boldsymbol{v}}[\mathrm{Reg}(\mathcal{M}; (\mathbf{0}, A_2, \ldots, A_n), \boldsymbol{v})] &\geq \frac{3}{1900} \sum_{i \in I} p_i x_i \\ &\geq \frac{3}{3800}\, b. \end{aligned} \tag{18}$$

This inequality holds for every algorithm $\mathcal{M}$, including randomized algorithms. Since the worst-case regret of an algorithm is at least its regret averaged over any prior,

$$\inf_{\mathcal{M}} \sup_{\boldsymbol{v}} \operatorname{Reg}(\mathcal{M}; \boldsymbol{v}) \geq \inf_{\mathcal{M}} \mathbb{E}_{\boldsymbol{v}}[\operatorname{Reg}(\mathcal{M}; \boldsymbol{v})] \geq \frac{3b}{3800}.$$

Thus no separate invocation of a finite-game version of Yao's principle is needed.

Removing the attainment assumption

It remains to justify the argument when the supremum defining b is not attained. Fix $\eta > 0$, and choose a feasible vector $\boldsymbol{x}^\eta$ whose objective value F_η satisfies

$$F_\eta \geq b - \eta.$$

Write $x_i = x_i^\eta$, $p_i = p_i(x_i)$, and again set

$$I_\eta := \{i : x_i \geq b\}.$$

Then

$$\sum_{i \in I_\eta} p_i x_i \geq F_\eta - b \sum_{i \notin I_\eta} p_i \geq \frac{b}{2} - \eta. \tag{19}$$

Replacing x_i by $7x_i/5$ remains feasible. Since the new objective is at most b,

$$F_\eta - x_i p_i + \frac{7}{5} x_i \, p_i\left(\frac{7}{5} x_i\right) \leq b.$$

It follows that

$$\frac{7}{5} x_i \, p_i\left(\frac{7}{5} x_i\right) \leq x_i p_i + \eta$$

and hence

$$\mathbb{P}\left[a_i \in \left[\theta_i + x_i, \theta_i + \frac{7}{5} x_i\right)\right] \geq \frac{2}{7} p_i - \frac{5\eta}{7x_i}. \tag{20}$$

Thus the common-component coefficient in (13) is replaced by

$$\left(\frac{p_i}{7} - \frac{5\eta}{14x_i}\right)_+.$$

The pigeonhole coefficient $3/380$ is unchanged.

If either easy binary case occurs, the preceding binary argument already proves the theorem. Otherwise, repeating (17) gives

$$\begin{aligned}
\mathbb{E}_{\boldsymbol{v}}[\operatorname{Reg}(\mathcal{M}; \boldsymbol{v})] &\geq \frac{3}{1900} \sum_{i \in I_\eta} p_i x_i - \sum_{i \in I_\eta} \frac{\eta}{28} \\
&\geq \frac{3}{1900} \left(\frac{b}{2} - \eta\right) - \frac{n\eta}{28}.
\end{aligned}$$

The total error is finite and tends to zero with η. Letting $\eta \downarrow 0$ proves

$$\operatorname{OPT}(\mathbf{0}, A_2, \ldots, A_n) \geq \frac{3b}{3800}$$

without assuming attainment. $\square$

Remark (Atoms and ties). The interval partitions above may be taken half-open, so atoms are assigned to exactly one interval. The convolution bounds are statements holding almost everywhere with respect to Lebesgue measure; endpoints of the common uniform intervals therefore do not affect the measure domination.

For the two binary cases, an atom may lie exactly on the decision boundary of Offset_θ. Fix the algorithm's tie-breaking rule and perturb the tested intrinsic value by ρ in the appropriate direction, making the relevant comparison strict. The resulting regret differs by at most $O(\rho)$. Taking the appropriate one-sided limit as $\rho \downarrow 0$ recovers the displayed non-strict bounds. This is the same perturbation convention used for atomic distributions in the binary analysis of [2].

6 Proof of the main theorem

Combining the linearization estimate (4) with Theorem 2 gives

$$\begin{aligned}
\mathrm{Reg}^{\uparrow}(\mathrm{Offset}_\theta; (\mathbf{0}, A_2, \ldots, A_n)) &\le \frac{51}{20}\, b \\
&\le \frac{51}{20} \cdot \frac{3800}{3}\ \mathrm{OPT}(\mathbf{0}, A_2, \ldots, A_n) \\
&= 3230\ \mathrm{OPT}(\mathbf{0}, A_2, \ldots, A_n). \qquad (21)
\end{aligned}$$

The comparisons between the reduced optimal regrets and the original optimal regret have the following directions:

$$\mathrm{OPT}(\mathbf{0}, A_2, \ldots, A_n) \le \mathrm{OPT}(A_1, A_2, \ldots, A_n), \qquad (22)$$

$$\mathrm{OPT}(A_1, \mathbf{0}) \le \mathrm{OPT}(A_1, A_2, \ldots, A_n). \qquad (23)$$

For (22), an algorithm for the anchored problem can generate an independent sample from A_1 and thereby simulate an algorithm for the original instance. For (23), one first simulates the second noise and then uses the binary embedding from Section 6.1 of [2]; the remaining items may be sent to arbitrarily negative intrinsic values.

Using (1), (2), (21), (22), and (23), we conclude

$$\begin{aligned}
\mathrm{Reg}(\mathrm{Offset}_\theta; \boldsymbol{A}) &\le 2\,\mathrm{Reg}^{\uparrow}(\mathrm{Offset}_\theta; (\mathbf{0}, A_2, \ldots, A_n)) + 2\,\mathrm{Reg}(\mathrm{Offset}_\theta; (A_1, \mathbf{0})) \\
&\le 2 \cdot 3230\ \mathrm{OPT}(\boldsymbol{A}) + 2 \cdot 24\ \mathrm{OPT}(\boldsymbol{A}) \\
&= 6508\ \mathrm{OPT}(\boldsymbol{A}).
\end{aligned}$$

This proves Theorem 1.

Corollary 1. *If all the noise distributions are symmetric, the greedy algorithm that selects the largest observed value is a* 6508*-approximation to the optimal regret.*

Proof. For a symmetric distribution A_i, the offset prescribed in [2] is $\theta(A_i) = 0$. Thus Offset_θ coincides with the greedy algorithm, and the claim follows from Theorem 1. □

7 Discussion

Theorem 1 improves the previously published constant from 57000 to 6508, thereby giving an affirmative answer to the corresponding open question in [2]. The result does not determine the optimal approximation factor: the strongest known lower-bound example has ratio 2.

The numerical improvement is concentrated in the anchored lower bound. The original proof partitions an interval into much narrower pieces and obtains a common-component mass of 1/550. Using intervals of length $2x_i/5$ gives mass 3/190, while the corresponding typical and atypical observation distributions contain a common uniform component with weights $p_i/7$ and 3/380. Balancing these two indistinguishable cases yields

$$\mathrm{OPT}(\mathbf{0}, A_2, \ldots, A_n) \geq \frac{3}{3800} b.$$

The rest of the argument uses the existing 24-approximation for binary instances and the existing general-to-anchored reduction.

References

[1] Mohammad Mahdian, Jieming Mao, Kangning Wang. *Regret Minimization with Noisy Observations*. Economics and Computation (EC), 2023. https://doi.org/10.1145/3580507.3597789

[2] Mohammad Mahdian, Jieming Mao, and Kangning Wang. Regret minimization with noisy observations. *arXiv preprint arXiv:2207.09435*, 2022. https://arxiv.org/abs/2207.09435.